

Serenify: AI-Driven Audio-Visual Therapy

1-line description

A web application merging Generative AI, music therapy, and interactive fluid simulations to promote mental well-being through multi-sensory synchronization between music, visuals, and emotions.

Description

Serenify is a "digital sanctuary" designed for users seeking relaxation, emotional regulation, or empathetic guidance. The experience is not passive but orchestrates three interconnected components driven by neuroaesthetics:

1. **AI Counseling & The "Musicologist"**: An empathic chatbot (Google Gemini) analyzes the user's sentiment to offer support and curate a specific musical selection based on the **ISO Principle**. The core innovation lies in the secondary agent, the **"AI Musicologist"**, which leverages LLM knowledge to extract technical metadata (Musical Key, BPM) from the chosen tracks, bypassing traditional API limitations.
2. **Evolutionary Visualizer**: A generative background built in **p5.js** simulating a "Primordial Soup" of organic particles. The system is **alive**: it reads the Musical Key (e.g., C Major) to harmonize collision sounds and uses the track's BPM to drive the fluid physics (velocity and turbulence), creating a visual dance perfectly synchronized with the audio.
3. **Therapeutic Rituals**:
 - *Breathe Mode*: A guided breathing interface based on **L-Systems** (recursive fractals) that expand and contract with the user's breath.
 - *Burn & Reframe*: A digital catharsis ritual where negative thoughts are visually incinerated (via pixel manipulation) and subsequently transformed by the AI into motivational phrases (**Cognitive Reframing**).

Challenges, accomplishments, and lessons learned

- **Challenges**: The major technical challenge was the "Black Box" of audio analysis: due to the deprecation of Spotify's Audio Features, we couldn't retrieve Key and BPM programmatically. We solved this by engineering a **Prompt Engineering pipeline** where Gemini acts as a music theory expert to retrieve this data in real-time.
- **Accomplishments**: We are proud to have achieved **Full Sensory Synchronisation**: the visuals are not a video loop, but a physics simulation reacting in real-time to the "sadness" (minor keys, slow particles) or "energy" (high BPM, turbulence) of the music. Furthermore, the integration of *Cognitive Reframing* closes the therapeutic loop, turning the destruction of a negative thought into positive construction.
- **Lessons learned**: We learned that LLMs can replace static databases for domain knowledge (such as musicology). We also specialize our understanding of graphics optimization in **p5.js**, managing the lifecycle (Object Pooling) of hundreds of interactive particles within a complex React architecture without performance drops.

Technology

- **Frontend:** React.js, p5.js (Creative Coding), CSS3 (Glassmorphism/Animations).
- **Backend:** Node.js, Express.js.
- **Cloud:** Firebase, Render.
- **AI & Data:** Google Gemini (2.0 Flash) for NLP, Sentiment Analysis, and Musicology; Spotify Web API for playback and search.
- **Audio:** Web Audio API (Procedural synthesis, Oscillators), Spotify Web Playback SDK.
- **Concepts:** Generative Art, Particle Systems, Perlin Noise, L-Systems (Fractals), Agent-based simulation.

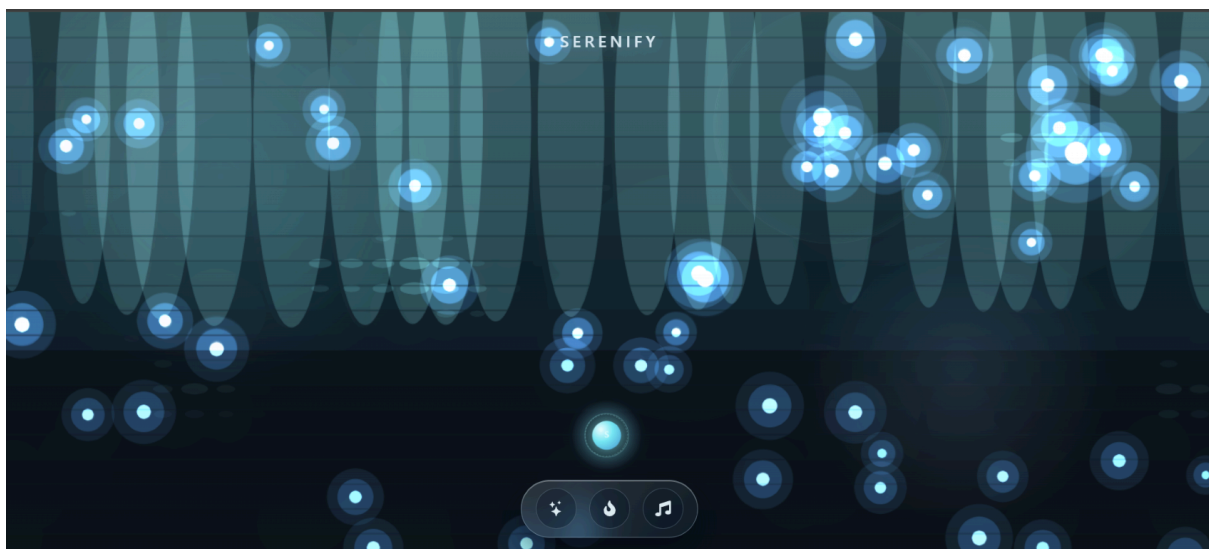
Students

- **Alberto Bollino:** Managed the Backend architecture and the "AI Musicologist" integration, handling prompt engineering logic and caching to optimize Gemini API calls. Handled the cloud architecture.
- **Wilma Bertilsson:** Focused on graphics, slides and presentation of the project.
- **Matteo Orlandin:** Developed the React Frontend interface, the UX (Glassmorphism), the Chat component, and the visual implementation of the "Burn" ritual.

Links

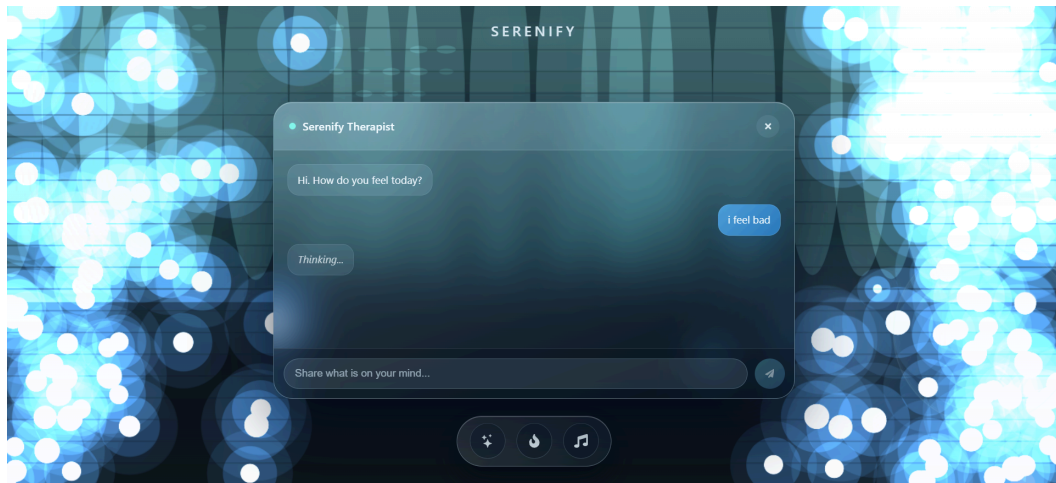
- **GitHub Repo:** <https://github.com/Quentin50ino/music-therapy-app>
- **Web App Demo:** <https://music-therapy-app-246ba.web.app/>
- **Video Demo:** https://www.youtube.com/watch?v=vko_JChvME4
- **Presentation:**
https://docs.google.com/presentation/d/1gS86u_rbCEXoV-6_YU0lcXCI1wVqut_pdi_nQkMutal/edit?usp=sharing

1 thumbnail image related to the projects

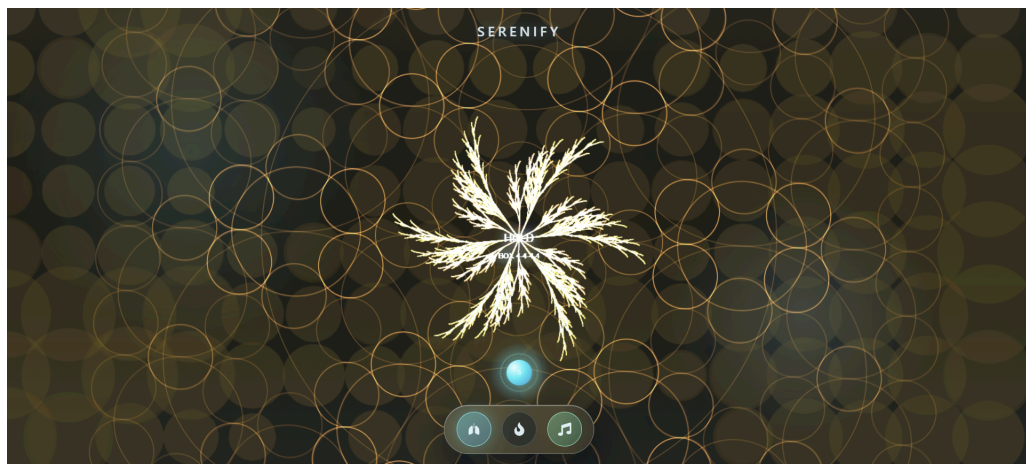


Pictures

1. **Flow Mode & Chat:** Screenshot showing the main interface with colored particles in the background and the side chat open with a musical suggestion from the AI.



2. **Breathe Mode:** Screenshot of the fractal mandala (L-System) at the moment of maximum expansion during inhalation.



3. **Burn & Reframe Ritual:** A split or single screenshot showing the moment the text is burning (fire effect) or the subsequent overlay with the 4 AI-generated reframing options.

